COHEN-MACAULAYNESS READ OFF FROM THE VANISHING OF EXT INTO THE NORMALIZED DUALIZING COMPLEX

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The degrees in which Ext into the normalized dualizing complex can be nonzero. Invoked at the start of the counted proof (source lines 3323-3324). Confines the Ext modules to a single relevant degree and fixes the comparison with the depth.
- The double dual statement identifying a finite module with the internal hom of its Ext into the dualizing complex. Invoked inside the counted proof (source lines 3331-3335). Turns a single nonzero Ext into an identification of the module as an Ext of the resulting module.
- The bound of the depth of a finite module by the dimension of its support. Invoked inside the counted proof (source lines 3338-3339). Supplies the reverse inequality that upgrades the single-nonzero condition to Cohen-Macaulayness.

Source reference. Original reference algebra-lemma-bound-depth: sources/stacks-dualizing/pinned-original-source.tex, source lines 3339–3339. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-bound-depth, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-dualizing: sources/stacks-dualizing/pinned-original-source.tex, source lines 2818–2818. Navigation pointer only: the counted text refers to the Stacks item lemma-dualizing, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

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Source reference. Original lemma lemma-sitting-in-degrees: sources/stacks-dualizing/pinned-original-source.tex, source lines 3232–3232. Navigation pointer only: the counted text refers to the Stacks item lemma-sitting-in-degrees, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a Noetherian local ring with normalized dualizing complex $\omega_A^\bullet$. Let M be a finite A-module. The following are equivalent*

(1) *M is Cohen-Macaulay,*
(2) *$\mathrm{Ext}_A^i(M, \omega_A^\bullet)$ is nonzero for at most one i,*
(3) *$\mathrm{Ext}_A^{-i}(M, \omega_A^\bullet)$ is zero for $i \neq \dim(\mathit{Supp}(M))$.*

Denote CM_d the category of finite Cohen-Macaulay A-modules of depth d. Then $M \mapsto \mathrm{Ext}_A^{-d}(M, \omega_A^\bullet)$ defines an anti-auto-equivalence of CM_d.

Proof. We will use the results of Lemma lemma-sitting-in-degrees without further mention. Fix a finite module M. If M is Cohen-Macaulay, then only $\mathrm{Ext}_A^{-d}(M, \omega_A^\bullet)$ can be nonzero, hence (1) $\Rightarrow$ (3). The implication (3) $\Rightarrow$ (2) is immediate. Assume (2) and let $N = \mathrm{Ext}_A^{-\delta}(M, \omega_A^\bullet)$ be the nonzero Ext where $\delta = \mathrm{depth}(M)$. Then, since

$$M[0] = R\mathrm{Hom}_A(R\mathrm{Hom}_A(M, \omega_A^\bullet), \omega_A^\bullet) = R\mathrm{Hom}_A(N[\delta], \omega_A^\bullet)$$

(Lemma lemma-dualizing) we conclude that $M = \mathrm{Ext}_A^{-\delta}(N, \omega_A^\bullet)$. Thus $\delta \geq \dim(\mathrm{Supp}(M))$. However, since we also know that $\delta \leq \dim(\mathrm{Supp}(M))$ (Algebra, Lemma algebra-lemma-bound-depth) we conclude that M is Cohen-Macaulay.

To prove the final statement, it suffices to show that $N = \mathrm{Ext}_A^{-d}(M, \omega_A^\bullet)$ is in CM_d for M in CM_d. Above we have seen that $M[0] = R\mathrm{Hom}_A(N[d], \omega_A^\bullet)$ and this proves the desired result by the equivalence of (1) and (3). $\square$